Chanyoung Park

CONTACT INFORMATION

Associate Professor
Data Science & Artificial Intelligence Lab (DSAIL)
Dept. of Industrial & Systems Engineering, KAIST
The Kim Jaechul Graduate School of AI, KAIST
Graduate School of Data Science, KAIST
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Google Scholar

CURRENT RESEARCH

- **AI for Scientific Discovery**
  - Materials, Chemistry
  - Biology, Engineering, EDA
- **Agentic AI**
- **LLM / Multimodal Data Mining**
  - Recommender System, Graph Neural Networks
  - LLM Safety, Model Efficiency, Video Language Model
  - Scene Graph Generation, Anomaly Detection, Time Series Analysis
- **Data-Centric AI**
  - Model Robustness, Adversarial Attack/Defense
  - Imbalanced Learning, Few-shot Learning, Explainability

POSITIONS

KAIST, Daejeon, South Korea — Mar 2025 – Present
- Associate Professor, Dept. of Industrial & Systems Engineering
- Associate Professor, The Kim Jaechul Graduate School of AI
- Associate Professor, Graduate School of Data Science

KAIST, Daejeon, South Korea — Nov 2020 – Feb 2025
- Assistant Professor, Dept. of Industrial & Systems Engineering
- Assistant Professor, The Kim Jaechul Graduate School of AI
- Assistant Professor, Graduate School of Data Science

University of Illinois at Urbana-Champaign, Urbana, IL, USA — Jan 2019 – Oct 2020
- Postdoctoral Reserach Fellow in Computer Science Department
  - Host: Prof. Jiawei Han
  - Topics: Multimodal data mining, Heterogeneous network representation learning

Microsoft Research, Beijing, China — Sep 2017 – Dec 2017
- Research Intern
  - Mentor: Dr. Xing Xie
  - Project: Collaborative Translational Metric Learning

NAVER, Seongnam, South Korea — Mar 2017 – Aug 2017
- Research Intern
  - Mentors: Dr. Jung-Tae Lee, and Dr. Min-Chul Yang
  - Project: Click-aware Purchase Prediction with Push at the Top

EDUCATION

Pohang University of Science and Technology, Pohang, South Korea — Feb 2019
- Ph.D. in Computer Science and Engineering
  - Thesis: Recommendation Framework via Matrix Factorization and Translation
  - Topics: Multimodal data mining, User behavior modeling, Recommender system

Sogang University, Seoul, South Korea — Feb 2014
- B.S. in Computer Science and Engineering

AWARDS & SCHOLARSHIPS

NVIDIA Academic Grant — 2026
- NVIDIA

Best Paper Award — 2026
- KDD 2026 Workshop on Personal Intelligence in Agentic AI Era, Jeju, South Korea

Best Paper Award — 2026

- KDD 2026 Workshop on Reliable Scientific Foundation Models, Jeju, South Korea

Google Research Award 2026
- Google

Impact Research Award (Achievement Award) 2025
- KAIST

Best Academic Award 2025
- Dept. of Industrial and Systems Engineering, KAIST

Song-Am Distinguished Research Award 2024
- College of Engineering, KAIST

Best Academic Award 2024
- Dept. of Industrial and Systems Engineering, KAIST

Best Paper Award 2024
- KDD 2024 Workshop on Federated Learning for Data Mining and Graph Analytics (FedKDD), Barcelona, Spain

Best Paper Award 2024
- KDD 2024 Workshop on Human-Interpretable AI, Barcelona, Spain

Gold Prize, 30th Samsung Humantech Paper Award 2024

Technology Innovation Awards 2023
- College of Engineering, KAIST

Top Reviewer of NeurIPS 2023

Best Paper Award 2023
- ICML 2023 Workshop on Computational Biology, Hawaii, USA

KAIST Excellence in Teaching Award of Year 2022 2023

Best Teaching Award 2023
- Dept. of Industrial and Systems Engineering, KAIST

ICDM Travel Award 2018
- International Conference on Data Mining, Singapore

Award of excellence 2017
- Microsoft Research Asia Internship Program, Beijing, China
- Awarded to the best intern

Naver Ph.D Fellowship 2016
- Awarded to outstanding Ph.D students majoring in computer science in Korea

RecSys Challenge 2015 2015
- Ranked 10 out of 850 teams from 49 countries (Top 1.1%)
- Task: Click/Purchase Prediction

Dean's List Fall 2013
- Sogang University

PUBLICATIONS

CONFERENCES

[C103] GeoGater: Meta-Feature Guided Adaptive Routing for Worldwide Image Geolocalization
Hoyoung Choi, Sungwon Kim, Jiwan Kim, **Chanyoung Park**
The 20th ACM International Conference on Web Search and Data Mining **(WSDM 2027)**

[C102] Personalized Visual Representation Alignment for Generative Multimodal Recommendation
Seonwoo Choi, Kibum Kim, **Chanyoung Park**, Donghyun Kim
IEEE/CVF Winter Conference on Applications of Computer Vision **(WACV 2027)**

[C101] Complex-Valued Periodic Stochastic Processes for Retrieving Unknown Atomic Structures
Gyoung S. Na, **Chanyoung Park**
The Fortieth Annual Conference on Neural Information Processing Systems **(NeurIPS 2026)**

[C100] Self-EvolveRec: Self-Evolving Recommender Systems with LLM-based Directional Feedback
Sein Kim, Sangwu Park, HongSeok Kang, Wonjoong Kim, Jimin Seo, Yeonjun In, Kanghoon Yoon, Hyunsik Jeon, **Chanyoung Park**
The Fortieth Annual Conference on Neural Information Processing Systems **(NeurIPS 2026)**

[C99] PAIR: Prefix-Aware Internal Reward Model for Multi-Turn Agent Optimization
Wonjoong Kim, Yeonjun In, Sangwu Park, Dongha Lee, **Chanyoung Park**
The Fortieth Annual Conference on Neural Information Processing Systems **(NeurIPS 2026)**

[C98] Chasing Label Shifters: A Change-Aware Framework for Dynamic Graph Node Classification
Junghoon Kim, Seungyoon Choi, Hyunsung Kim, **Chanyoung Park**
The Fortieth Annual Conference on Neural Information Processing Systems **(NeurIPS 2026)**

[C97] MSP-LLM: A Structured Large Language Model Framework for Inorganic Material Synthesis Planning
Heewoong Noh, Gyoung S. Na, Namkyeong Lee, **Chanyoung Park**
The Fortieth Annual Conference on Neural Information Processing Systems **(NeurIPS 2026)**

[C96] Personalize-then-Store: Benchmarking and Learning Personalized Memory for Long-Horizon Agents
Yeonjun In, Wonjoong Kim, Sangwu Park, Kanghoon Yoon, **Chanyoung Park**
The Fortieth Annual Conference on Neural Information Processing Systems **(NeurIPS 2026)**

[C95] Screening Lipid Nanoparticles through Structure-Ratio Alignment
Yoonho Lee, Yunhak Oh, Hoyoung Choi, **Chanyoung Park**
The Fortieth Annual Conference on Neural Information Processing Systems **(NeurIPS 2026)**

[C94] scTrilemma: Balancing Identity, Invariance, and Fidelity in Single-Cell Representation Learning
Yunhak Oh, Yoonho Lee, Junseok Lee, Namkyeong Lee, Sang-Yeon Hwang, Yinhua Piao, Hyomin Kim, Seonghwan Kim, Jaechang Lim, Woo Youn Kim, Sungsoo Ahn, **Chanyoung Park**
The Fortieth Annual Conference on Neural Information Processing Systems **(NeurIPS 2026)**

[C93] AutoSaddler: Automatic Harness Optimization with Durable Updates from Agent Execution Traces
Sungho Park, Wonjoong Kim, Rongyuan Tan, Jue Zhang, Wook-Shin Han, Pengfei Gao, **Chanyoung Park**, Qingwei Lin, Saravan Rajmohan, Dongmei Zhang
The Fortieth Annual Conference on Neural Information Processing Systems **(NeurIPS 2026)**

[C92] Compositional Spectral Prompts for LLM-based Online Time Series Forecasting
Seungyoon Choi, Hyunchul Kim, Jae-Gil Lee, **Chanyoung Park**
ACM International Conference on Information and Knowledge Management **(CIKM 2026)**

[C91] ELASTIC: Trajectory-Based Synchronization of Event and Tracking Data in Soccer
Hyunsung Kim, Hoyoung Choi, Kunhee Lee, Sangwoo Seo, Tom Boomstra, Jinsung Yoon, **Chanyoung Park**
ACM International Conference on Information and Knowledge Management **(CIKM 2026)**

[C90] DefaultGNN: A Dual-Perspective GNN Framework for Predicting Corporate Default from Buyer-Seller Transaction Networks
Junghoon Kim, Hyunsung Kim, Seungyoon Choi, Kyoungyong Park, Jihun Lee, Yonggu Ji, **Chanyoung Park**
ACM International Conference on Information and Knowledge Management **(CIKM 2026) (Applied Research Track)**

[C89] A Behavioral Trait Leaks into Preferences: Diagnosing Trait Interference in LLM User Simulators
Chaehyun Kim, Sein Kim, Hongseok Kang, **Chanyoung Park**
ACM International Conference on Information and Knowledge Management **(CIKM 2026) (Short Paper)**

[C88] HUGETIME at eRisk 2026: Bucket-wise Evidence Pruning for Contextualized Early Depression Detection
Ilho Shin, Hyunchul Kim, **Chanyoung Park**
The Conference and Labs of the Evaluation Forum **(CLEF 2026)**
Ranked 1st in Task 2: Contextualized Early Detection of Depression at CLEF eRisk 2026

[C87] Sink-Token-Aware Pruning for Fine-Grained Video Understanding in Efficient Video LLMs
Kibum Kim, Jiwan Kim, Kyle Min, Yueqi Wang, Jinyoung Moon, Julian McAuley, **Chanyoung Park**
European Conference on Computer Vision **(ECCV 2026)**

[C86] Why and When Visual Token Pruning Fails? A Study on Relevant Visual Information Shift in MLLMs Decoding
Jiwan Kim, Kibum Kim, Wonjoong Kim, Byung-Kwan Lee, **Chanyoung Park**
European Conference on Computer Vision **(ECCV 2026)**

[C85] PathCRF: Ball-Free Soccer Event Detection via Possession Path Inference from Player Trajectories
Hyunsung Kim, Kunhee Lee, Sangwoo Seo, Sang-Ki Ko, Jinsung Yoon, **Chanyoung Park**
ACM SIGKDD Conference on Knowledge Discovery and Data Mining **(KDD 2026)**

[C84] Learning Probabilistic Compositional Representation of Crystalline Materials
Namkyeong Lee, Heewoong Noh, Gyoung S. Na, Jimeng Sun, Tianfan Fu, Marinka Zitnik, **Chanyoung Park**
ACM SIGKDD Conference on Knowledge Discovery and Data Mining **(KDD 2026) (AI for Science Track))**

[C83] EqGINO: Equivariant Geometry-Informed Fourier Neural Operators for 3D PDEs
Sungwon Kim, Juho Song, Seungmin Shin, Guimok Cho, Sangkook Kim, **Chanyoung Park**
International Conference on Machine Learning **(ICML 2026)**

[C82] Beyond the Final Answer: Evaluating the Reasoning Trajectories of Tool-Augmented Agents
Wonjoong Kim, Sangwu Park, Yeonjun In, Sein Kim, Dongha Lee, **Chanyoung Park**
International Conference on Machine Learning **(ICML 2026)**

[C81] Learning Adaptive Perturbation-Conditioned Contexts for Robust Transcriptional Response Prediction
Yinhua Piao, Hyomin Kim, Seonghwan Kim, Yunhak Oh, Junhyeok Jeon, Sang-Yeon Hwang, Jaechang Lim, Woo Youn Kim, **Chanyoung Park**, Sungsoo Ahn
International Conference on Machine Learning **(ICML 2026)**

[C80] Time-PEFT: Temporal and Multichannel Complexity-Based Fine-Tuning for Time-Series Foundation Models
Jihye Na, Patara Trirat, **Chanyoung Park**, Jae-Gil Lee
International Conference on Machine Learning **(ICML 2026)**

[C79] SelfJudge: Faster Speculative Decoding via Self-Supervised Judge Verification
Kanghoon Yoon, Minsub Kim, Sungjae Lee, Joonhyung Lee, Sunghyeon Woo, Yeonjun In, Se Jung Kwon, **Chanyoung Park**, Dongsoo Lee
International Conference on Machine Learning **(ICML 2026)**

[C78] Reasoning Structure Matters for Safety Alignment of Reasoning Models
Yeonjun In, Wonjoong Kim, Sangwu Park, **Chanyoung Park**
The 64th Annual Meeting of the Association for Computational Linguistics **(ACL 2026)**

[C77] IR-Agent: Expert-Inspired LLM Agents for Structure Elucidation from Infrared Spectra
Heewoong Noh, Namkyeong Lee, Gyoung S. Na, Kibum Kim, **Chanyoung Park**
The Fourteenth International Conference on Learning Representations **(ICLR 2026)**

[C76] CompoDistill: Attention Distillation for Compositional Reasoning in Multimodal LLMs
Jiwan Kim, Kibum Kim, Sangwoo Seo, **Chanyoung Park**
The Fourteenth International Conference on Learning Representations **(ICLR 2026)**

[C75] Token-Efficient Item Representation via Images for LLM Recommender Systems
Kibum Kim, Sein Kim, HongSeok Kang, Jiwan Kim, Heewoong Noh, Yeonjun In, Kanghoon Yoon, Jinoh Oh, Julian McAuley, **Chanyoung Park**
The Fourteenth International Conference on Learning Representations **(ICLR 2026)**

[C74] PULSE: Socially-Aware User Representation Modeling Toward Parameter-Efficient Graph Collaborative Filtering
Doyun Choi*, Cheonwoo Lee*, Biniyam Aschalew Tolera, Taewook Ham, **Chanyoung Park**, Jaemin Yoo
The Web Conference **(WWW 2026)**

[C73] RAG-Enhanced Collaborative LLM Agents for Drug Discovery
Namkyeong Lee, Edward De Brouwer, Ehsan Hajiramezanali, Tommaso Biancalani, **Chanyoung Park**, Gabriele Scalia
The 40th Annual AAAI Conference on Artificial Intelligence **(AAAI 2026)**

[C72] 3D Interaction Geometric Pre-training for Molecular Relational Learning
Namkyeong Lee, Yunhak Oh, Heewoong Noh, Gyoung S. Na, Minkai Xu, Hanchen Wang, Tianfan Fu, **Chanyoung Park**
The Thirty-Ninth Annual Conference on Neural Information Processing Systems **(NeurIPS 2025 (Spotlight))**

[C71] Disentangling Hyperedges through the Lens of Category Theory
Yoonho Lee, Junseok Lee, Sangwoo Seo, Sungwon Kim, Yeongmin Kim, **Chanyoung Park**
The Thirty-Ninth Annual Conference on Neural Information Processing Systems **(NeurIPS 2025)**

[C70] Training Robust Graph Neural Networks by Modeling Noise Dependencies
Yeonjun In, Kanghoon Yoon, Sukwon Yun, Kibum Kim, Sungchul Kim, **Chanyoung Park**
The Thirty-Ninth Annual Conference on Neural Information Processing Systems **(NeurIPS 2025)**

[C69] Is Safety Standard Same for Everyone? User-Specific Safety Evaluation of Large Language Models
Yeonjun In, Wonjoong Kim, Kanghoon Yoon, Sungchul Kim, Mehrab Tanjim, Kibum Kim, **Chanyoung Park**
Conference on Empirical Methods in Natural Language Processing **(EMNLP 2025 Findings)**

[C68] Disambiguation in Conversational Question Answering in the Era of LLM: A Survey
Md Mehrab Tanjim*, Yeonjun In*, Xiang Chen, Victor S. Bursztyn, Ryan A. Rossi, Sungchul Kim, Guang-Jie Ren, Vaishnavi Muppala, Shun Jiang, Yongsung Kim, **Chanyoung Park**
Conference on Empirical Methods in Natural Language Processing **(EMNLP 2025)**

[C67] Target Circuit Matching in Large-Scale Netlists using GNN-Based Region Prediction
Sangwoo Seo, Jimin Seo, Yoonho Lee, Donghyeon Kim, Hyejin Shin, Banghyun Sung, **Chanyoung Park**
International Conference on Computer-Aided Design **(ICCAD 2025)**

[C66] Lost in Sequence: Do Large Language Models Understand Sequential Recommendation?
Sein Kim*, HongSeok Kang*, Kibum Kim, Jiwan Kim, Donghyun Kim, Min-Chul Yang, Kwangjin Oh, Julian McAuley, **Chanyoung Park**
ACM SIGKDD Conference on Knowledge Discovery and Data Mining **(KDD 2025)**

[C65] Oldie but Goodie: Re-illuminating Label Propagation on Graphs with Partially Observed Features
Sukwon Yun, Xin Liu, Yunhak Oh, Junseok Lee, Tianlong Chen, Tsuyoshi Murata, **Chanyoung Park**
ACM SIGKDD Conference on Knowledge Discovery and Data Mining **(KDD 2025)**

[C64] Electron-Informed Coarse-Graining Molecular Representation Learning for Real-World Molecular Physics
Gyoung S. Na, **Chanyoung Park**
ACM SIGKDD Conference on Knowledge Discovery and Data Mining **(KDD 2025)**

[C63] On Measuring Unnoticeability of Graph Adversarial Attacks: Observations, New Measure, and Applications
Hyeonsoo Jo*, Hyunjin Hwang*, Fanchen Bu, Soo Yong Lee, **Chanyoung Park**, Kijung Shin
ACM SIGKDD Conference on Knowledge Discovery and Data Mining **(KDD 2025)**

[C62] Thickness-aware E(3)-Equivariant 3D Mesh Neural Networks
Sungwon Kim, Namkyeong Lee, Yunyoung Doh, Seungmin Shin, Guimok Cho, Seung-Won Jeon, Sangkook Kim, **Chanyoung Park**
International Conference on Machine Learning (**ICML 2025**)

[C61] Global Context-aware Representation Learning for Spatially Resolved Transcriptomics
Yunhak Oh*, Junseok Lee*, Yeongmin Kim, Sangwoo Seo, Namkyeong Lee, **Chanyoung Park**
International Conference on Machine Learning (**ICML 2025**)

[C60] Disentangling and Generating Modalities for Recommendation in Missing Modality Scenarios
Jiwan Kim, Hongseok Kang, Sein Kim, Kibum Kim, **Chanyoung Park**
ACM SIGIR Conference on Research and Development in Information Retrieval (**SIGIR 2025**)

[C59] Dynamic Time-aware Continual User Representation Learning
Seungyoon Choi, Sein Kim, Hongseok Kang, Wonjoong Kim, **Chanyoung Park**
ACM SIGIR Conference on Research and Development in Information Retrieval (**SIGIR 2025**)

[C58] Subgraph Federated Learning for Local Generalization
Sungwon Kim, Yoonho Lee, Yunhak Oh, Namkyeong Lee, Sukwon Yun, Junseok Lee, Sein Kim, Carl Yang, **Chanyoung Park**
The Thirteenth International Conference on Learning Representations **(ICLR 2025 (Oral))**

[C57] Weakly Supervised Video Scene Graph Generation via Natural Language Supervision
Kibum Kim, Kanghoon Yoon, Yeonjun In, Jaehyeong Jeon, Jinyoung Moon, Donghyun Kim, **Chanyoung Park**
The Thirteenth International Conference on Learning Representations **(ICLR 2025)**

[C56] Self-Supervised Diffusion Models for Electron-Aware Molecular Representation Learning
Gyoung S. Na, **Chanyoung Park**
The Thirteenth International Conference on Learning Representations **(ICLR 2025)**

[C55] Diversify-verify-adapt: Efficient and Robust Retrieval-Augmented Ambiguous Question Answering
Yeonjun In, Sungchul Kim, Ryan A. Rossi, Mehrab Tanjim, Tong Yu, Ritwik Sinha, **Chanyoung Park**
Annual Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics **(NAACL 2025)**

[C54] SIMPLOT: Enhancing Chart Question Answering by Distilling Essentials
Wonjoong Kim*, Sangwu Park*, Yeonjun In, Seokwon Han, **Chanyoung Park**
Annual Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics **(NAACL 2025 Findings)**

[C53] RA-SGG: Retrieval-Augmented Scene Graph Generation Framework via Multi-Prototype Learning
Kanghoon Yoon, Kibum Kim, Jaehyeong Jeon, Yeonjun In, Donghyun Kim, **Chanyoung Park**
Thirty-Nineth AAAI Conference on Artificial Intelligence **(AAAI 2025)**

[C52] Revisiting Fake News Detection: Towards Temporality-aware Evaluation by Leveraging Engagement Earliness
Junghoon Kim*, Junmo Lee*, Yeonjun In, Kanghoon Yoon, **Chanyoung Park**
The 18th ACM International Conference on Web Search and Data Mining **(WSDM 2025 (Oral))**

[C51] Retrieval-Retro: Retrieval-based Inorganic Retrosynthesis with Expert Knowledge
Heewoong Noh, Namkyeong Lee, Gyoung S. Na, **Chanyoung Park**
Thirty-eightth Conference on Neural Information Processing Systems **(NeurIPS 2024)**

[C50] Debiased Graph Poisoning Attack via Contrastive Surrogate Objective
Kanghoon Yoon, Yeonjun In, Namkyeong Lee, Kibum Kim, **Chanyoung Park**
ACM International Conference on Information and Knowledge Management **(CIKM 2024)**

[C49] Vision Language Model is NOT All You Need: Augmentation Strategies for Molecule Language Models
Namkyeong Lee, Siddhartha Laghuvarapu, **Chanyoung Park**, Jimeng Sun
ACM International Conference on Information and Knowledge Management **(CIKM 2024)**

[C48] Semantic Diversity-aware Prototype-based Learning for Unbiased Scene Graph Generation
Jaehyeong Jeon, Kibum Kim, Kanghoon Yoon, **Chanyoung Park**
European Conference on Computer Vision (**ECCV 2024**)

[C47] Multiplexed Immunofluorescence Image Analysis through an Efficient Multiplex Network
Sukwon Yun, Jie Peng, Alexandro Trevino, **Chanyoung Park**, Tianlong Chen
European Conference on Computer Vision (**ECCV 2024**)

[C46] Large Language Models meet Collaborative Filtering: An Efficient All-round LLM-based Recommender System
Sein Kim*, Hongseok Kang*, Seungyoon Choi, Donghyun Kim, Minchul Yang, **Chanyoung Park**
ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD 2024**)

[C45] Self-Explainable Temporal Graph Networks based on Graph Information Bottleneck
Sangwoo Seo, Sungwon Kim, Jihyeong Jung, Yoonho Lee, **Chanyoung Park**
ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD 2024**)

[C44] Unsupervised Episode Generation for Graph Meta-learning
Jihyeong Jung, Sangwoo Seo, Sungwon Kim, **Chanyoung Park**
International Conference on Machine Learning (**ICML 2024**)

[C43] LLM4SGG: Large Language Models for Weakly Supervised Scene Graph Generation
Kibum Kim, Kanghoon Yoon, Jaehyeong Jeon, Yeonjun In, Jinyoung Moon, Donghyun Kim, **Chanyoung Park**
IEEE/CVF Computer Vision and Pattern Recognition Conference (**CVPR 2024**)

[C42] Self-guided Robust Graph Structure Refinement
Yeonjun In, Kanghoon Yoon, Kibum Kim, Kijung Shin, **Chanyoung Park**
The Web Conference 2024 (**WWW 2024 (Oral)**)

[C41] DSLR: Diversity Enhancement and Structure Learning for Rehearsal-based Graph Continual Learning
Seungyoon Choi*, Wonjoong Kim*, Sungwon Kim, Yeonjun In, Sein Kim, **Chanyoung Park**
The Web Conference 2024 (**WWW 2024 (Oral)**)

[C40] Adaptive Self-training Framework for Fine-grained Scene Graph Generation
Kibum Kim*, Kanghoon Yoon*, Yeonjun In, Jinyoung Moon, Donghyun Kim, **Chanyoung Park**
Twelfth International Conference on Learning Representations (**ICLR 2024**)

[C39] Sterling: Synergistic Representation Learning on Bipartite Graphs
Baoyu Jing, Yuchen Yan, Kaize Ding, **Chanyoung Park**, Yada Zhu, Huan Liu, Hanghang Tong
Thirty-Eighth AAAI Conference on Artificial Intelligence (**AAAI 2024 (Oral)**)

[C38] Interpretable Prototype-based Graph Information Bottleneck
Sangwoo Seo, Sungwon Kim, **Chanyoung Park**
Thirty-seventh Conference on Neural Information Processing Systems (**NeurIPS 2023**) **(Gold Prize, 30th Samsung Humantech Paper Award)**

[C37] Density of States Prediction of Crystalline Materials via Prompt-guided Multi-Modal Transformer
Namkyeong Lee*, Heewoong Noh*, Sungwon Kim, Dongmin Hyun, Gyoung S. Na, **Chanyoung Park** Thirty-seventh Conference on Neural Information Processing Systems (**NeurIPS 2023**)

[C36] MUSE: Music Recommender System with Shuffle Play Recommendation Enhancement
Yunhak Oh*, Sukwon Yun*, Dongmin Hyun, Sein Kim, **Chanyoung Park**
ACM International Conference on Information and Knowledge Management **(CIKM 2023)**

[C35] Class Label-aware Graph Anomaly Detection
Junghoon Kim, Yeonjun In, Kanghoon Yoon, Junmo Lee, **Chanyoung Park**
ACM International Conference on Information and Knowledge Management **(CIKM 2023)** (Short Paper)

[C34] Structural Mixup for Graph Neural Networks
Junghurn Kim*, Sukwon Yun*, **Chanyoung Park**
ACM International Conference on Information and Knowledge Management **(CIKM 2023)** (Short Paper)

[C33] Similarity Preserving Adversarial Graph Contrastive Learning
Yeonjun In*, Kanghoon Yoon*, **Chanyoung Park**
ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD 2023**)

[C32] Shift-Robust Molecular Relational Learning with Causal Substructure
Namkyeong Lee, Kanghoon Yoon, Gyoung S. Na, Sein Kim, **Chanyoung Park**
ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD 2023**)

[C31] Task-Equivariant Graph Few-shot Learning
Sungwon Kim, Junseok Lee, Namkyeong Lee, Wonjoong Kim, Seungyoon Choi, **Chanyoung Park**
ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD 2023**)

[C30] Task Relation-aware Continual User Representation Learning
Sein Kim, Namkyeong Lee, Donghyun Kim, Min-Chul Yang, **Chanyoung Park**
ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD 2023**)

[C29] Conditional Graph Information Bottleneck for Molecular Relational Learning
Namkyeong Lee, Dongmin Hyun, Gyoung S. Na, Sungwon Kim, Junseok Lee, **Chanyoung Park**
International Conference on Machine Learning (**ICML 2023**)

[C28] MELT: Mutual Enhancement of Long-Tailed User and Item for Sequential Recommendation
Kibum Kim, Dongmin Hyun, Sukwon Yun, **Chanyoung Park**
ACM SIGIR Conference on Research and Development in Information Retrieval (**SIGIR 2023**)

[C27] Unbiased Heterogeneous Scene Graph Generation with Relation-aware Message Passing Neural Network
Kanghoon Yoon*, Kibum Kim*, Jinyoung Moon, **Chanyoung Park**
Thirty-Seventh AAAI Conference on Artificial Intelligence (**AAAI 2023**)

[C26] Heterogeneous Graph Learning for Multi-modal Medical Data Analysis
Sein Kim, Namkyeong Lee, Junseok Lee, Dongmin Hyun, **Chanyoung Park**
Thirty-Seventh AAAI Conference on Artificial Intelligence (**AAAI 2023 (Oral)**)

[C25] Generating Multiple-Length Summaries via Reinforcement Learning for Unsupervised Sentence Summarization
Dongmin Hyun, Xiting Wang, **Chanyoung Park**, Xing Xie, Hwanjo Yu
Conference on Empirical Methods in Natural Language Processing (**EMNLP 2022 Findings**)

[C24] Set2Box: Similarity Preserving Representation Learning for Sets
Geon Lee, **Chanyoung Park**, Kijung Shin
The IEEE International Conference on Data Mining (**ICDM 2022**)

[C23] LTE4G: Long-Tail Experts for Graph Neural Networks
Sukwon Yun, Kibum Kim, Kanghoon Yoon, **Chanyoung Park**
ACM International Conference on Information and Knowledge (**CIKM 2022**)

[C22] Relational Self-Supervised Learning on Graphs
Namkyeong Lee, Dongmin Hyun, Junseok Lee, **Chanyoung Park**
ACM International Conference on Information and Knowledge (**CIKM 2022**)

[C21] Beyond Learning from Next Item: Sequential Recommendation via Personalized Interest Sustainability
Dongmin Hyun, **Chanyoung Park**, Junsu Cho, Hwanjo Yu
ACM International Conference on Information and Knowledge (**CIKM 2022**)

[C20] Nonlinearity Encoding for Extrapolation of Neural Networks
Gyoung S. Na, **Chanyoung Park**
ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD 2022**)

[C19] GraFN: Semi-Supervised Node Classification on Graph with Few Labels via Non-Parametric Distribution Assignment
Junseok Lee, Yunhak Oh, Yeonjun In, Namkyeong Lee, Dongmin Hyun, **Chanyoung Park**
ACM SIGIR Conference on Research and Development in Information Retrieval (**SIGIR 2022**) (Short Paper)

[C18] AHP: Learning to Negative Sample for Hyperedge Prediction
Hyunjin Hwang, Seungwoo Lee, **Chanyoung Park**, Kijung Shin
ACM SIGIR Conference on Research and Development in Information Retrieval (**SIGIR 2022**) (Short Paper)

[C17] Augmentation-Free Self-Supervised Learning on Graphs
Namkyeong Lee, Junseok Lee, **Chanyoung Park**
Thirty-Sixth AAAI Conference on Artificial Intelligence (**AAAI 2022**)

[C16] Learnable Structural Semantic Readout for Graph Classification
Dongha Lee, Su Kim, Seonghyeon Lee, **Chanyoung Park**, Hwanjo Yu
IEEE International Conference on Data Mining (**ICDM 2021**) (Short Paper)

[C15] Bootstrapping User and Item Representations for One-Class Collaborative Filtering
Dongha Lee, Seongku Kang, Hyunjun Ju, **Chanyoung Park**, Hwanjo Yu
ACM SIGIR Conference on Research and Development in Information Retrieval (**SIGIR 2021**)

[C14] HDMI: High-order Deep Multiplex Infomax
Baoyu Jing, **Chanyoung Park**, Hanghang Tong
The Web Conference (**WWW 2021**)

[C13] Interest Sustainability-Aware Recommender System
Dongmin Hyun, Junsu Cho, **Chanyoung Park**, Hwanjo Yu
IEEE International Conference on Data Mining (**ICDM 2020**)

[C12] Unsupervised Differentiable Multi-aspect Network Embedding
Chanyoung Park, Carl Yang, Qi Zhu, Donghyun Kim, Hwanjo Yu and Jiawei Han
ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD 2020**)

[C11] Unsupervised Attributed Multiplex Network Embedding
Chanyoung Park, Donghyun Kim, Jiawei Han, Hwanjo Yu
Thirty-Fourth AAAI Conference on Artificial Intelligence (**AAAI 2020**)

[C10] Task-Guided Pair Embedding in Heterogeneous Network
Chanyoung Park, Donghyun Kim, Qi Zhu, Jiawei Han, Hwanjo Yu
ACM International Conference on Information and Knowledge (**CIKM 2019**)

[C9] BHIN2vec: Balancing the Type of Relation in Heterogeneous Information Network
Seonghyun Lee, **Chanyoung Park**, Hwanjo Yu
ACM International Conference on Information and Knowledge (**CIKM 2019**)

[C8] Action Space Learning for Heterogeneous User Behavior Prediction
Dongha Lee, **Chanyoung Park**, Hyunjun Ju, Junyoung Hwang, Hwanjo Yu
International Joint Conference on Artificial Intelligence (**IJCAI 2019**)

[C7] Sequential and Diverse Recommendation with Long Tail
Yejin Kim, Kwangseob Kim, **Chanyoung Park**, Hwanjo Yu
International Joint Conference on Artificial Intelligence (**IJCAI 2019**)

[C6] Collaborative Translational Metric Learning
Chanyoung Park, Donghyun Kim, Xing Xie, Hwanjo Yu
IEEE International Conference on Data Mining (**ICDM 2018**)

[C5] Review Sentiment-Guided Scalable Deep Recommender System
Dongmin Hyun, **Chanyoung Park**, Min-Chul Yang, Ilhyeon Song, Jung-Tae Lee, Hwanjo Yu
ACM SIGIR conference on Research and Development in Information Retrieval (**SIGIR 2018)** (Short Paper)

[C4] Do "Also-Viewed" Products Help User Rating Prediction?
Chanyoung Park, Donghyun Kim, Jinoh Oh, Hwanjo Yu
International World-Wide Web Conference (**WWW 2017**)

[C3] Convolutional Matrix Factorization for Document Context-Aware Recommendation
Donghyun Kim, **Chanyoung Park**, Jinoh Oh, Hwanjo Yu
ACM international conference on Recommender System (**RecSys 2016**)

[C2] TRecSo: Enhancing Top-k Recommendation With Social Information
Chanyoung Park, Donghyun Kim, Jinoh Oh, Hwanjo Yu
International World-Wide Web Conference (**WWW 2016**) (Poster)

[C1] Predicting User Purchase in E-commerce by Comprehensive Feature Engineering and Decision Boundary Focused Under-Sampling
Chanyoung Park, Donghyun Kim, Jinoh Oh, Hwanjo Yu
ACM international conference on Recommender System Challenge (**ACM RecSysChallenge 2015**, 10th place, top 1.1%)

JOURNALS

[J14] A New DAWN for Fake News Detection: Leveraging Engagement Earliness for Temporality-aware Evaluation
Junghoon Kim*, Junmo Lee*, Yeonjun In, Kanghoon Yoon, **Chanyoung Park**
ACM Transactions on Intelligent Systems and Technology (2026)

[J13] Physics-Embedded Graph Neural Operator for Interaction-Controlled Colloidal Aggregation
Yongjoon Choe*, Sungwon Kim*, Susan E. Burns, **Chanyoung Park**
Water Research (2026)

[J12] Single-cell RNA Sequencing Data Imputation Using Bi-level Feature Propagation
Junseok Lee*, Sukwon Yun*, Yeongmin Kim, Tianlong Chen, Manolis Kellis, **Chanyoung Park**
Briefings in Bioinformatics (2024)

[J11] Deep Single-cell RNA-seq Data Clustering with Graph Prototypical Contrastive Learning
Junseok Lee, Sungwon Kim, Dongmin Hyun, Namkyeong Lee, Yejin Kim, **Chanyoung Park**
Bioinformatics (2023)

[J10] Self-Supervised Graph Representation Learning via Positive Mining
Namkyeong Lee, Junseok Lee, **Chanyoung Park**
Information Sciences (2022.9)

[J9] Learning to Utilize Auxiliary Reviews for Recommendation
Dongmin Hyun, **Chanyoung Park**, Junsu Cho, Hwanjo Yu
Information Sciences (2021.2)

[J8] Deep Multiplex Graph Infomax: Attentive Multiplex Network Embedding using Global Information
Chanyoung Park, Jiawei Han, Hwanjo Yu
Knowledge-Based Systems (2020.6)

[J7] Improved prediction of miRNA–disease associations based on matrix completion with network regularization
Jihwan Ha, Chihyun Park, **Chanyoung Park**, Sanghyun Park
Cells (2020.4) (SCIE)

[J6] Click-aware Purchase Prediction with Push at the Top
Chanyoung Park, Donghyun Kim, Min-Chul Yang, Jung-Tae Lee, Hwanjo Yu
Information Sciences (2020.6)

[J5] Inferring miRNA-disease Interactions using Probabilistic Matrix Factorization
Jihwan Ha, Chihyun Park, **Chanyoung Park**, Sanghyun Park
Journal of Biomedical Informatics (2020.2)

[J4] An Encoder-Decoder Switch Network for Purchase Prediction
Chanyoung Park, Donghyun Kim, Hwanjo Yu
Knowledge-Based Systems (2019.12)

[J3] Target-aware Convolutional Neural Network for Target-level Sentiment Analysis
Dongmin Hyun, **Chanyoung Park**, Min-Chul Yang, Ilhyeon Song, Jung-Tae Lee, Hwanjo Yu
Information Sciences (2019.7)

[J2] Deep Hybrid Recommender Systems via Exploiting Document Context and Statistics of Items
Donghyun Kim, **Chanyoung Park**, Jinoh Oh, Hwanjo Yu
Information Sciences (2017.11)

[J1] Improving top-K recommendation with truster and trustee relationship in user trust network
Chanyoung Park, Donghyun Kim, Jinoh Oh, Hwanjo Yu
Information Sciences (2016.12)

WORKSHOPS

[W33] scTrilemma: Balancing Identity, Invariance, and Fidelity in Single-Cell Representation Learning
Yunhak Oh, Yoonho Lee, Junseok Lee, Namkyeong Lee, Sang-Yeon Hwang, Yinhua Piao, Hyomin Kim, Seonghwan Kim, Jaechang Lim, Woo Youn Kim, Sungsoo Ahn, **Chanyoung Park**
The Third Workshop on GenAI for Health: Agentic Systems, Clinical Trust, and Future Potential **(NeurIPS 2026 Workshop)**

[W32] Machine Collective Intelligence with Canonical Syntax Representations for Explainable Scientific Discovery
Gyoung S. Na, **Chanyoung Park**
Workshop on AI for Science: Verification in the Age of AI Scientists **(NeurIPS 2026 Workshop)**

[W31] Bridging Prediction and Planning: From Synthesis Temperature Distributions to Optimization-Guided Experiments
Jaewon Bae, Seungyun Shin, **Chanyoung Park**
Workshop on AI for Accelerated Materials Design (AI4Mat) **(NeurIPS 2026 Workshop)**

[W30] Combining Physical and Retrieved Empirical Priors for Solid-State Synthesis Condition Prediction
Woojin Bae, Hyunsung Jang, Jaewon Bae, **Chanyoung Park**
Workshop on AI for Accelerated Materials Design (AI4Mat) **(NeurIPS 2026 Workshop)**

[W29] A Structure-to-Density Model for Trajectory-Free Lithium Transport Screening
Kunmin Jang, Dongin Kim, Woojin Bae, Jaewon Bae, **Chanyoung Park**
Workshop on AI for Accelerated Materials Design (AI4Mat) **(NeurIPS 2026 Workshop)**

[W28] Tokenizing Antibody–Antigen Complex Structures via Energy-Based Surrogates
Junseok Lee, Yunhak Oh, Namkyeong Lee, Hyunchul Kim, **Chanyoung Park**
Workshop on Reliable Scientific Foundation Models **(KDD 2026 Workshop)**

[W27] Bridging Circuit Graphs and Language: A Cross-Modal Transformer for Netlist-to-Text Alignment and Generation
Sangwoo Seo, Hyunsung Kim, Hyunchul Kim, Byeonggon Kang, Rex Ying, **Chanyoung Park**
Workshop on Frontiers in Graph Machine Learning for the Large Model Era **(KDD 2026 Workshop)**

[W26] Adaptive Graph Rewiring to Mitigate Over-Squashing in Mesh-Based GNNs for Fluid Dynamics Simulations
Sangwoo Seo, Hyunsung Kim, Jiwan Kim, **Chanyoung Park**
Workshop on Reliable Scientific Foundation Models **(KDD 2026 Workshop)**

[W25] A Structured LLM Framework for Inorganic Material Synthesis Planning
Heewoong Noh, Gyoung S. Na, Namkyeong Lee, **Chanyoung Park**
Workshop on Agentic AI for Scientific and Societal Advances **(KDD 2026 Workshop)**

[W24] Beyond the Final Answer: Evaluating the Reasoning Trajectories of Tool-Augmented Agents
Wonjoong Kim, Sangwu Park, Yeonjun In, Sein Kim, Dongha Lee, **Chanyoung Park**
Workshop on Evaluation and Trustworthiness of Agentic AI **(KDD 2026 Workshop)**

[W23] Compositional Spectral Prompts for LLM-based Online Time Series Forecasting
Seungyoon Choi, Hyunchul Kim, Jae-Gil Lee, **Chanyoung Park**
Workshop on Mining and Learning from Time Series **(KDD 2026 Workshop)**

[W22] Personalize-then-Store: Benchmarking and Learning Personalized Memory for Long-Horizon Agents
Yeonjun In, Wonjoong Kim, Sangwu Park, Kanghoon Yoon, **Chanyoung Park**
Workshop on Personal Intelligence in Agentic AI Era **(KDD 2026 Workshop)**

[W21] Rethinking Failure Attribution in Multi-Agent Systems: A Multi-Perspective Benchmark and Evaluation
Yeonjun In, Mehrab Tanjim, Jayakumar Subramanian, Sungchul Kim, Uttaran Bhattacharya, Wonjoong Kim, Sangwu Park, Somdeb Sarkhel, **Chanyoung Park**
Workshop on Agentic Software Engineering **(KDD 2026 Workshop)**

[W20] Self-EvolveRec: Self-Evolving Recommender Systems with LLM-based Directional Feedback
Sein Kim, Sangwu Park, HongSeok Kang, Wonjoong Kim, Jimin Seo, Yeonjun In, Kanghoon Yoon, Hyunsik Jeon, **Chanyoung Park**
Workshop on Agentic Software Engineering **(KDD 2026 Workshop)**

[W19] Why and When Visual Token Pruning Fails? A Study on Relevant Visual Information Shift in MLLMs Decoding
Jiwan Kim, Kibum Kim, Wonjoong Kim, Byung-Kwan Lee, **Chanyoung Park**
The Fourth International Workshop on Resource-Efficient Learning for Knowledge Discovery **(KDD 2026 Workshop)**

[W18] Revisiting CF-Integrated LLM Recommenders through Dimensional Collapse
Jimin Seo, HongSeok Kang, Sein Kim, Jiwan Kim, **Chanyoung Park**
Geometric Space, Architecture and Learning Objectives for Large Pre-Trained Models **(KDD 2026 Workshop)**

[W17] Physics-Embedded Graph Neural Operator for Interaction-Controlled Colloidal Aggregation
Yongjoon Choe, Sungwon Kim, Susan E. Burns, **Chanyoung Park**
Workshop on Reliable Scientific Foundation Models **(KDD 2026 Workshop)**

[W16] EqGINO: Equivariant Geometry-Informed Fourier Neural Operators for 3D PDEs
Sungwon Kim, Juho Song, Seungmin Shin, Guimok Cho, Sangkook Kim, **Chanyoung Park**
ICLR 2026 Workshop on AI and Partial Differential Equations (Oral) (ICLR 2026 Workshop)

[W15] Self-EvolveRec: Self-Evolving Recommender Systems with LLM-based Directional Feedback
Sein Kim, Sangwu Park, HongSeok Kang, Wonjoong Kim, Jimin Seo, Yeonjun In, Kanghoon Yoon, **Chanyoung Park**
ICLR 2026 Workshop on AI with Recursive Self-Improvement (RSI 2026) (ICLR 2026 Workshop)

[W14] MSP-LLM: A Unified Large Language Model Framework for Complete Material Synthesis Planning
Heewoong Noh, Gyoung S. Na, Namkyeong Lee, **Chanyoung Park**
AI4Mat: AI for Accelerated Materials Design (ICLR 2026 Workshop)

[W13] Synergistic Multi-Task Learning for Electronic Density of States Prediction
Kunmin Jang, Dongik Park, Jaewon Bae, **Chanyoung Park**
AI4Mat: AI for Accelerated Materials Design (ICLR 2026 Workshop)

[W12] Learning k-Resolved Electronic Structure via Soft Energy Occupancy Prediction
Dongik Park, Kunmin Jang, Jaewon Bae, Dongin Kim, **Chanyoung Park**
AI4Mat: AI for Accelerated Materials Design (ICLR 2026 Workshop)

[W11] Learning Adaptive Perturbation-Conditioned Contexts for Robust Transcriptional Response Prediction
Yinhua Piao, Hyomin Kim, Seonghwan Kim, Yunhak Oh, Junhyeok Jeon, Sang-Yeon Hwang, Jaechang Lim, Woo Youn Kim, **Chanyoung Park**, Sungsoo Ahn
ICLR 2026 Workshop on Generative and Experimental Perspectives for Biomolecular Design (ICLR 2026 Workshop)

[W10] RAG-Enhanced Collaborative LLM Agents for Drug Discovery
Namkyeong Lee, Edward De Brouwer, Ehsan Hajiramezanali, **Chanyoung Park**, Gabriele Scalia
MLGenX Workshop at ICLR 2025 (MLGenX 2025 Spotlight)

[W9] 3D Interaction Geometric Pre-training for Molecular Relational Learning
Namkyeong Lee, Yunhak Oh, Heewoong Noh, Gyoung S. Na, Tianfan Fu, **Chanyoung Park**
NeurIPS 2024 Workshop on AI for New Drug Modalities (AIDrugX)

[W8] Subgraph Federated Learning for Local Generalization
Sungwon Kim, Yoonho Lee, Yunhak Oh, Namkyeong Lee, Sukwon Yun, Junseok Lee, Sein Kim, Carl Yang, **Chanyoung Park**
KDD 2024 Workshop on Human-Interpretable AI (Oral) **(Best Paper Award)**

[W7] Interpretable Graph Model with Prototype-Based Graph Information Bottleneck
Sangwoo Seo, Sungwon Kim, **Chanyoung Park**
KDD 2024 Workshop on Human-Interpretable AI (Oral) **(Best Paper Award)**

[W6] Stoichiometry Representation Learning with Polymorphic Crystal Structures
Namkyeong Lee, Heewoong Noh, Gyoung S. Na, Tianfan Fu, Jimeng Sun, **Chanyoung Park**
NeurIPS 2023 Workshop on AI for Scientific Discovery: From Theory to Practice (AI4Science)

[W5] Electron-Derived Molecular Representation Learning for Real-World Molecular Physics
Gyoung S. Na, **Chanyoung Park**
NeurIPS 2023 Workshop on AI for Scientific Discovery: From Theory to Practice (AI4Science)

[W4] Deep single-cell RNA-seq data clustering with graph prototypical contrastive learning
Junseok Lee, Sungwon Kim, Dongmin Hyun, Namkyeong Lee, Yejin Kim, **Chanyoung Park**
ICML 2023 Workshop on Computational Biology

[W3] Denoising single-cell RNA-seq data using Feature Propagation
Sukwon Yun, Junseok Lee, **Chanyoung Park**
ICML 2023 Workshop on Computational Biology (Oral) **(Best Paper Award)**

[W2] Predicting Density of States via Multi-modal Transformer
Namkyeong Lee, Heewoong Noh, Sungwon Kim, Dongmin Hyun, Gyoung S. Na, **Chanyoung Park**
ICLR 2023 Workshop on Machine Learning for Materials (ML4Materials)

[W1] Shift-Robust Node Classification via Graph Adversarial Clustering
Qi Zhu, Chao Zhang, **Chanyoung Park**, Carl Yang, Jiawei Han
NeurIPS 2022 GLFrontiers Workshop

PATENT

[P1] Method for context-aware recommendation by considering contextual information of document and apparatus for the same
Donghyun Kim, **Chanyoung Park**, Jinoh Oh, Hwanjo Yu
No. 10-18771610000, issued July 2018.

PREPRINT

[A5] Do VLAs Really Follow Instructions? Diagnosing How-Level Instruction Following in Vision-Language-Action Models
Sangwu Park, Wonjoong Kim, Yeonjun In, **Chanyoung Park**
arXiv preprint **(arXiv:2610.21251)**

[A4] CondMSP: Condition-Informed and Resource-Aware Inorganic Materials Synthesis Planning
Heewoong Noh, Gyoung S. Na, Elton Pan, Namkyeong Lee, **Chanyoung Park**
arXiv preprint **(arXiv:2610.15603)**

[A3] ManiPhysicsBench: Physics-Based Assessment of Object Preservation in VLA Manipulation
Sangwu Park, Yeonjun In, Wonjoong Kim, Sungwon Kim, Sein Kim, **Chanyoung Park**
arXiv preprint **(arXiv:2610.02802)**

[A2] ActiveSaddler: Automated Curriculum Learning for Agent Harness Optimization
Sungho Park, Wonjoong Kim, Jue Zhang, Wook-Shin Han, Pengfei Gao, **Chanyoung Park**, Yongqiang Yao, Rao Fu, Elsie Nallipogu, Qingwei Lin, Victor Rühle
arXiv preprint **(arXiv:2610.00906)**

[A1] Adaptive Graph Rewiring to Mitigate Over-Squashing in Mesh-Based GNNs for Fluid Dynamics Simulations
Sangwoo Seo, Hyunsung Kim, Jiwan Kim, **Chanyoung Park**

TEACHING

Instructor

- IE343: Statistical Machine Learning, KAIST — 2021S, 2022S, 2023S, 2024S, 2026S
- DS535 (KSE801): Recommender System and Machine Learning on Graphs, KAIST — 2021F, 2022F, 2023F, 2024F
- CoE202: Basics of Artificial Intelligence, KAIST — 2021F
- DS503: Machine Learning for Data Science (Deep Learning), KAIST — 2022S,2023S,2024S

Teaching Assistant

- CSED352: Data Communications, POSTECH — 2014F
- CSED312: Operating System, POSTECH — 2016F

PROFESSIONAL SERVICES

AREA CHAIR / SENIOR PROGRAM COMMITTEE

- The International Conference on Machine Learning (ICML) [2025-present]
- The International Conference on Learning Representations (ICLR) [2025-present]
- ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD) [2025-present]
- Conference on Neural Information Processing Systems (NeurIPS) [2024-present]
- AAAI Conference on Artificial Intelligence (AAAI) [2026-present]
- ICML 2024 AI4Science Workshop [2024-present]
- NeurIPS Workshop (AI for Science: from Theory to Practice) [2023-present]

EVENT ORGANIZATIONS

- Organizer, KDD'26 Workshop on SciSoc Agents & LLMs [2026]
- General Chair, The ACM International Conference on Information and Knowledge Management (CIKM) [2025]
- Proceedings Chair, The ACM International Conference on Information and Knowledge Management (CIKM) [2023]

JOURNAL EDITORSHIP

- Action Editor in Transactions on Machine Learning Research (TMLR)
- Frontiers in Artificial Intelligence (Graph representation learning: Methods and Applications)
- Frontiers in Big Data (Reviews in Big Data Networks: 2022)